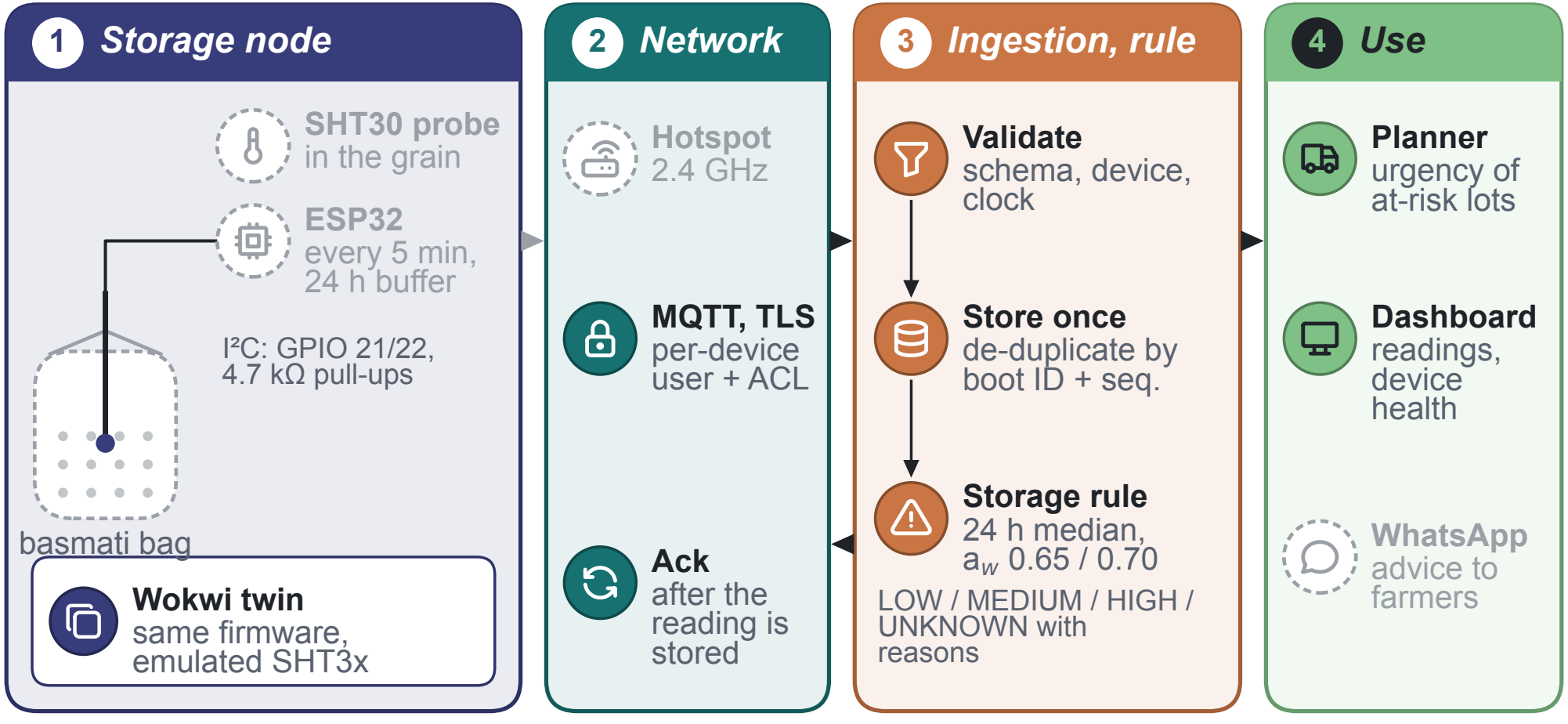

1 Storage node
SHT30 probe
in the grain
ESP32
every 5 min,
24 h buffer
I²C: GPIO 21/22,
4.7 kΩ pull-ups
basmati bag
Wokwi twin
same firmware,
emulated SHT3x
2 Network
Hotspot
2.4 GHz
MQTT, TLS
per-device
user + ACL
Ack
after the
reading is
stored
3 Ingestion, rule
Validate
schema, device,
clock
Store once
de-duplicate by
boot ID + seq.
Storage rule
24 h median,
a_w 0.65 / 0.70
LOW / MEDIUM / HIGH /
UNKNOWN with
reasons
4 Use
Planner
urgency of
at-risk lots
Dashboard
readings,
device
health
WhatsApp
advice to
farmers